

Q1.

Use g as 9.8 m s^{-2} in this question.

Dominic, a bungee jumper of mass 75 kg , has his ankles attached to one end of a cord. The other end of the cord is attached to a bridge which is 50 metres above the surface of a river.

The cord can be modelled as a light elastic cord of natural length 25 metres and modulus of elasticity 3200 N . Dominic is modelled as a particle.

Dominic steps off the bridge at the point where the cord is attached and falls vertically downwards.

- (a) Find Dominic's speed at the point when the cord initially becomes taut.

(2)

- (b) Determine whether or not Dominic enters the river and gets wet.

(5)

- (c) One limitation of this model is that Dominic is not a particle.

Explain the effect of revising this assumption on your answer to part (b).

(2)

(Total 9 marks)

Q2.

A cricket ball of mass 156 grams is thrown from a point which is 1.5 metres above the ground, with a speed of 12 m s^{-1} .

A tennis ball of mass 58 grams is also thrown from the same point, with the same speed.

Prove that both balls hit the ground with the same speed.

Clearly state any assumptions you have made and how you have used them.

(Total 5 marks)

Q3.

In this question use $g = 10 \text{ m s}^{-2}$.

A particle, of mass 2 kg , is attached to one end of a light elastic string of natural length 0.2 metres .

The other end of the string is attached to a fixed point O .

The particle is pulled down and released from rest at a point 0.6 metres directly below O .

The particle then moves vertically and next comes to rest when it is 0.1 metres below O .

Assume that no air resistance acts on the particle.

- (a) Find the maximum speed of the particle.

(6)

- (b) Describe one way in which the model you have used could be refined.

(1)

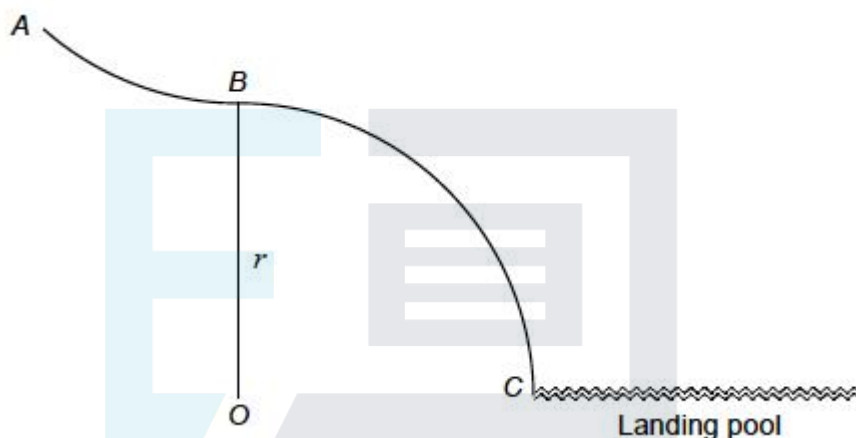
(Total 7 marks)

Q4.

The diagram shows part of a water park slide, ABC .

The slide is in the shape of two circular arcs, AB and BC , each of radius r .

The point A is at a height of $\frac{r}{4}$ above B .
 The circular arc BC has centre O and B is vertically above O .
 These points are joined as shown in the diagram.



A child starts from rest at A , moves along the slide past the point B and then loses contact with the slide at a point D .

The angle between the vertical, OB , and OD is θ

Assume that the slide is smooth.

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- (a) Show that the speed v of the child at D is given by $v = \sqrt{\frac{gr}{2}(5 - 4\cos\theta)}$, where g is the acceleration due to gravity.

(3)

- (b) Find θ , giving your answer to the nearest degree.

(2)

- (c) A refined model takes into account air resistance. Explain how taking air resistance into account would affect your answer to part (b).

(2)

- (d) In reality the slide is not smooth. It has a surface with the same coefficient of friction between the slide and the child for its entire length.

Explain why the frictional force experienced by the child is not constant.

(1)

(Total 8 marks)

Q5.

Tim is playing cricket. He hits a ball at a point A . The speed of the ball immediately after being hit is 11 m s^{-1} .

The ball strikes a tree at a point B . The height of B is 5 metres above the height of A .

The ball is to be modelled as a particle of mass 0.16 kg being acted upon only by gravity.

- (a) Calculate the initial kinetic energy of the ball. (2)
 - (b) Calculate the potential energy gained by the ball as it moves from the point A to the point B . (2)
 - (c) (i) Find the kinetic energy of the ball immediately before it strikes the tree. (2)
 - (ii) Hence find the speed of the ball immediately before it strikes the tree. (2)
- (Total 8 marks)**

Q6.

- (a) An elastic string has natural length l and modulus of elasticity λ . The string is stretched from length l to length $l + e$.

Show, by integration, that the work done in stretching the string is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg , is attached to one end of a light elastic string. The other end of the string is attached to a fixed point O .

The string has natural length 1.6 m and modulus of elasticity 392 N .

- (i) Find the extension of the string when the particle hangs in equilibrium. (2)
- (ii) The particle is pulled down to a point A , which is 2.2 m below the point O .
Calculate the elastic potential energy in the string. (3)
- (iii) The particle is released when it is at rest at the point A .
Calculate the distance of the particle from the point A when its speed first reaches 0.8 m s^{-1} . (5)

(Total 13 marks)

Q7.

Carol, a circus performer, is on a swing. She jumps off the swing and lands in a safety net. When Carol leaves the swing, she has a speed of 7 m s^{-1} and she is at a height of 8 metres above the safety net.

Carol is to be modelled as a particle of mass 52 kg being acted upon only by gravity.

- (a) Find the kinetic energy of Carol when she leaves the swing. (2)
 - (b) Show that the kinetic energy of Carol when she hits the net is 5350 J, correct to three significant figures. (3)
 - (c) Find the speed of Carol when she hits the net. (3)
- (Total 8 marks)**

Q8.

Two particles, *A* and *B*, are connected by a light elastic string that passes through a hole at a point *O* in a rough horizontal table. The edges of the hole are smooth. Particle *A* has a mass of 8 kg and particle *B* has a mass of 3 kg.

The elastic string has natural length 3 metres and modulus of elasticity 60 newtons.

Initially, particle *A* is held 3.5 metres from the point *O* on the surface of the table and particle *B* is held at a point 2 metres vertically below *O*.

The coefficient of friction between the table and particle *A* is 0.4.

The two particles are released from rest.

- (a) (i) Show that initially particle *A* moves towards the hole in the table. (3)
- (ii) Show that initially particle *B* also moves towards the hole in the table. (2)
- (b) Calculate the **initial** elastic potential energy in the string. (2)
- (c) Particle *A* comes permanently to rest when it has moved 0.46 metres, at which time particle *B* is still moving upwards.

Calculate the distance that particle *B* has moved when it is at rest for the first time.

(7)
(Total 14 marks)

Q9.

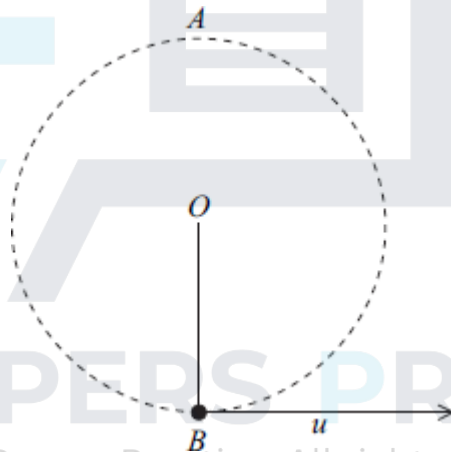
A plane is dropping packets of aid as it flies over a flooded village. The speed of a packet when it leaves the plane is 60 m s^{-1} . The packet has mass 25 kg.

The packet falls a vertical distance of 34 metres to reach the ground.

- (a) Calculate the kinetic energy of the packet when it leaves the plane. (2)
- (b) Calculate the potential energy lost by the packet as it falls to the ground. (2)
- (c) Assume that the effect of air resistance on the packet as it falls can be neglected.
- (i) Find the kinetic energy of the packet when it reaches the ground. (2)
- (ii) Hence find the speed of the packet when it reaches the ground. (2)
- (Total 8 marks)

Q10.

A small bead, of mass m , is suspended from a fixed point O by a light inextensible string of length a . With the string taut, the bead is at the point B , vertically below O , when it is set into vertical circular motion with an initial horizontal velocity u , as shown in the diagram.



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The string does not become slack in the subsequent motion. The velocity of the bead at the point A , where A is vertically above O , is v .

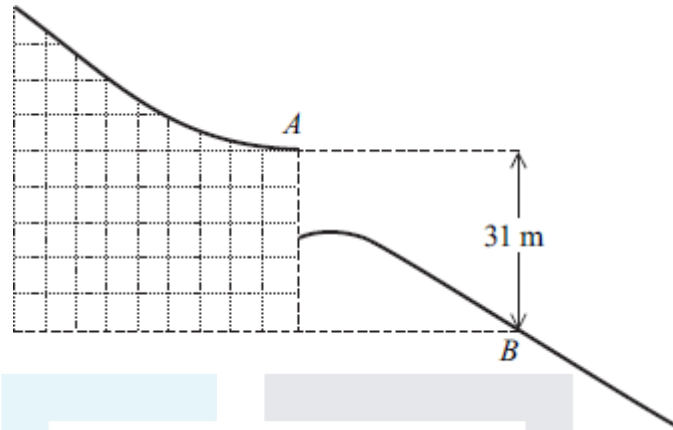
- (a) Show that $v^2 = u^2 - 4ag$. (2)
- (b) The ratio of the tensions in the string when the bead is at the two points A and B is $2 : 5$.
- (i) Find u in terms of g and a . (7)
- (ii) Find the ratio $u : v$. (2)
- (Total 11 marks)

Q11.

Alan, of mass 76 kg, performed a ski jump. He took off at the point A at the end of the ski run with a speed of 28 m s^{-1} and landed at the point B .

The level of the point B is 31 metres vertically below the level of the point A , as shown in the diagram.

Assume that his weight is the only force that acted on Alan during the jump.



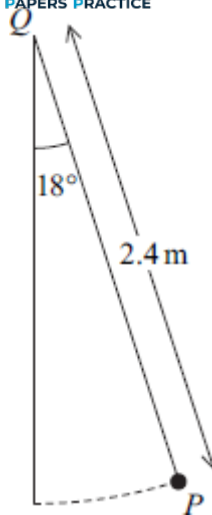
- (a) Calculate the kinetic energy of Alan when he was at the point A . (2)
- (b) Calculate the potential energy lost by Alan during the jump as he moved from the point A to the point B . (2)
- (c) (i) Find the kinetic energy of Alan when he reached the point B . (2)
- (ii) Hence find the speed of Alan when he reached the point B . (2)

(Total 8 marks)

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Q12.

Simon, a small child of mass 22 kg, is on a swing. He is swinging freely through an angle of 18° on both sides of the vertical. Model Simon as a particle, P , of mass 22 kg, attached to a fixed point, Q , by a light inextensible rope of length 2.4 m.



- (a) Find Simon's maximum speed as he swings.

(4)

- (b) Calculate the tension in the rope when Simon's speed is a maximum.

(3)

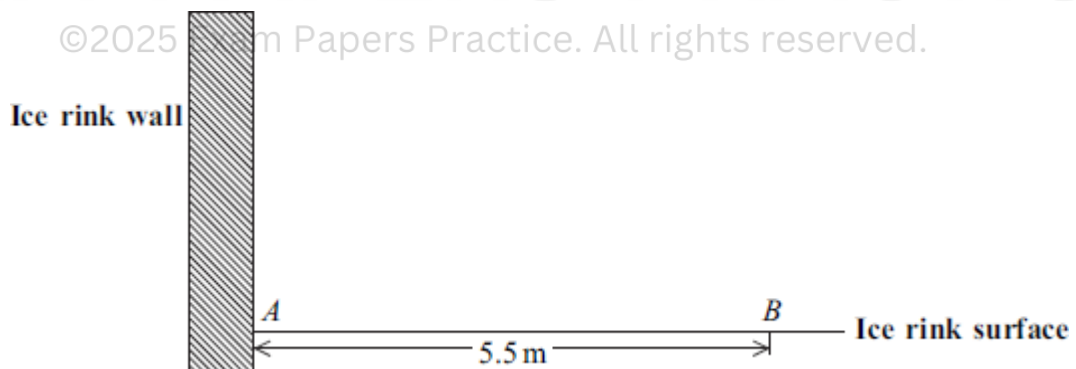
(Total 7 marks)

Q13.

Zoë carries out an experiment with a block, which she places on the horizontal surface of an ice rink. She attaches one end of a light elastic string to a fixed point, A , on a vertical wall at the edge of the ice rink at the height of the surface of the ice rink.

The block, of mass 0.4 kg , is attached to the other end of the string. The string has natural length 5 m and modulus of elasticity 120 N .

The block is modelled as a particle which is placed on the surface of the ice rink at a point B , where AB is perpendicular to the wall and of length 5.5 m .



The block is set into motion at the point B with speed 9 m s^{-1} directly towards the point A . The string remains horizontal throughout the motion.

- (a) Initially, Zoë assumes that the surface of the ice rink is smooth.

Using this assumption, find the speed of the block when it reaches the point A .

(4)

- (b) Zoë now assumes that friction acts on the block. The coefficient of friction between

the block and the surface of the ice rink is μ .

- (i) Find, in terms of g and μ , the speed of the block when it reaches the point A .

(6)

- (ii) The block rebounds from the wall in the direction of the point B . The speed of the block immediately after the rebound is half of the speed with which it hit the wall.

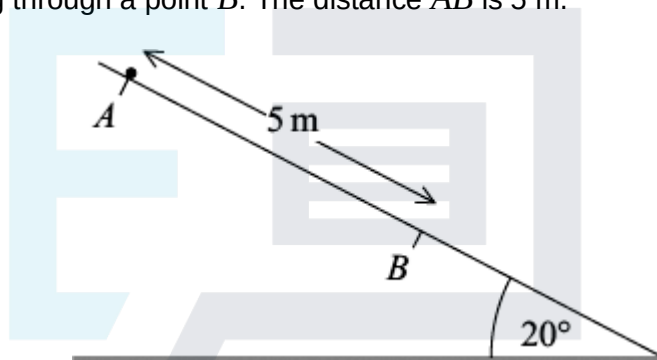
Find μ if the block comes to rest just as it reaches the point B .

(6)

(Total 16 marks)

Q14.

A particle is placed on a smooth plane which is inclined at an angle of 20° to the horizontal. The particle, of mass 4 kg , is released from rest at a point A and travels down the plane, passing through a point B . The distance AB is 5 m .



- (a) Find the potential energy lost as the particle moves from point A to point B .

(2)

- (b) Hence write down the kinetic energy of the particle when it reaches point B .

(1)

- (c) Hence find the speed of the particle when it reaches point B .

(2)

(Total 5 marks)

Q15.

A pump is being used to empty a flooded basement.

In one minute, 400 litres of water are pumped out of the basement.

The water is raised 8 metres and is ejected through a pipe at a speed of 2 m s^{-1} .

The mass of 400 litres of water is 400 kg .

- (a) Calculate the gain in potential energy of the 400 litres of water.

(1)

- (b) Calculate the gain in kinetic energy of the 400 litres of water.

(1)

- (c) Hence calculate the power of the pump, giving your answer in watts.

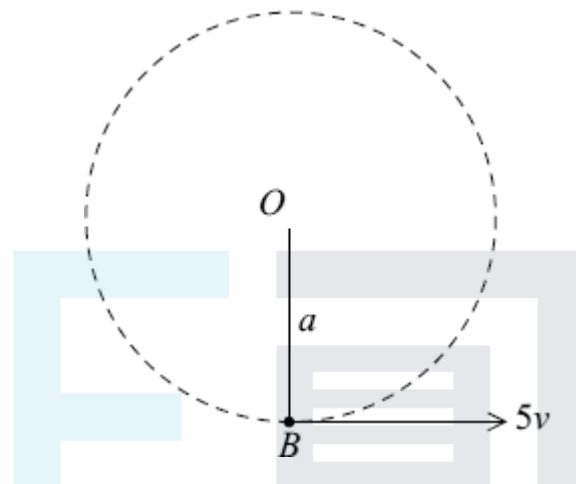
(2)

(Total 4 marks)

Q16.

A light inextensible string, of length a , has one end attached to a fixed point O .

A small bead, of mass m , is attached to the other end of the string. The bead is moving in a vertical circle, centre O . When the bead is at B , vertically below O , the string is taut and the bead is moving with speed $5v$.



- (a) The speed of the bead at the highest point of its path is $3v$.

Find v in terms of a and g .

(4)

- (b) Find the ratio of the greatest tension to the least tension in the string, as the bead travels around its circular path.

(5)

(Total 9 marks)

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Q17.

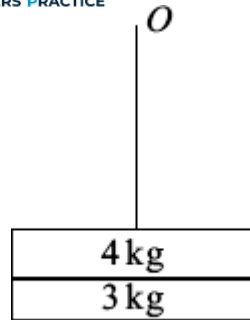
- (a) An elastic string has natural length l and modulus of elasticity λ . The string is stretched from length l to length $l + e$.

Show, by integration, that the work done in stretching the string is $\frac{\lambda e^2}{2l}$

(3)

- (b) A block, of mass 4 kg, is attached to one end of a light elastic string. The string has natural length 2 m and modulus of elasticity 196 N. The other end of the string is attached to a fixed point O .

- (i) A second block, of mass 3 kg, is attached to the 4 kg block and the system hangs in equilibrium, as shown in the diagram.



Find the extension in the string.

(3)

- (ii) The block of mass 3 kg becomes detached from the 4 kg block and falls to the ground. The 4 kg block now begins to move vertically upwards.

Find the extension of the string when the 4 kg block is next at rest.

(6)

- (iii) Find the extension of the string when the speed of the 4 kg block is a maximum.

(3)

(Total 15 marks)

Q18.

In an Olympic diving competition, Kim, who has mass 58 kg, dives from a fixed platform, 10 metres above the surface of the pool. She leaves the platform with a speed of 2 m s^{-1} .

Assume that Kim's weight is the only force that acts on her after she leaves the platform. Kim is to be modelled as a particle which is initially 1 metre above the platform.

- (a) Calculate Kim's initial kinetic energy.

(2)

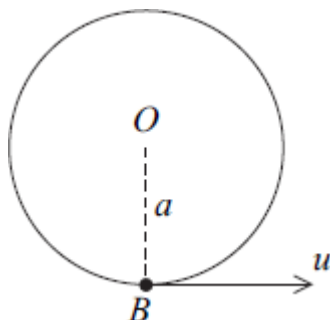
- (b) By using conservation of energy, find Kim's speed when she is 6 metres below the platform.

(5)

(Total 7 marks)

Q19.

A smooth wire is fixed in a vertical plane so that it forms a circle of radius a metres and centre O . A bead, B , of mass 0.3 kg, is threaded on the wire and is set in motion with a speed $u \text{ m s}^{-1}$ at the lowest point of its circular path, as shown in the diagram.

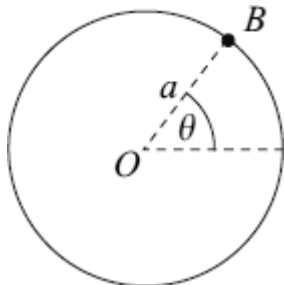


- (a) Show that, if the bead is going to make complete revolutions around the wire,

$$u > 2\sqrt{ag}$$

(3)

- (b) At time t seconds, the angle between OB and the horizontal is θ , as shown in the diagram.



It is given that $u = \sqrt{\frac{9}{2}ag}$.

- (i) Find the reaction of the bead on the wire, giving your answer in terms of g and θ .

(5)

- (ii) Find θ when this reaction is zero.

(2)

(Total 10 marks)

Q20.

At a theme park, a light elastic rope is used to bring a carriage to rest at the end of a ride.

The carriage has mass 200 kg and is travelling at 8 m s^{-1} when the elastic rope is attached to the carriage as it passes over a point O . The other end of the elastic rope is fixed to the point O . The carriage then moves along a horizontal surface until it is brought to rest. The elastic rope is then detached so that the carriage remains at rest.

The elastic rope has natural length 6 m and modulus of elasticity 1800 N. The rope, once taut, remains horizontal throughout the motion.

- (a) Calculate the elastic potential energy of the rope when the carriage is 10 m from O .

(3)

- (b) A student's simple model assumes that there are no resistance forces acting on the carriage so that it is brought to rest by the elastic rope alone.

Find the distance of the carriage from O when it is brought to rest.

(3)

- (c) The student improves the model by also including a constant resistance force of 800 N which acts while the carriage is in motion.

Find the distance of the carriage from O when it is brought to rest.

(8)
(Total 14 marks)

Q21.

A bungee jumper, of mass 49 kg, is attached to one end of a light elastic cord of natural length 22 metres and modulus of elasticity 1078 newtons. The other end of the cord is attached to a horizontal platform, which is at a height of 60 metres above the ground.

The bungee jumper steps off the platform at the point where the cord is attached, and falls vertically. The bungee jumper can be modelled as a particle. Assume that Hooke's Law applies whilst the cord is taut and that air resistance is negligible throughout the motion.

When the bungee jumper has fallen x metres, his speed is $v \text{ m s}^{-1}$.

- (a) By considering energy, show that, when x is greater than 22,

$$5v^2 = 318x - 5x^2 - 2420$$

(6)

- (b) Explain why x must be greater than 22 for the equation in part (a) to be valid.

(1)

- (c) Find the maximum value of x .

(4)

- (d) (i) Show that the speed of the bungee jumper is a maximum when $x = 31.8$.

(3)

- (ii) Hence find the maximum speed of the bungee jumper.

(2)

(Total 16 marks)

Q22.

John is at the top of a cliff, looking out over the sea. He throws a rock, of mass 3 kg, horizontally with a velocity of 4 m s^{-1} .

The rock falls a vertical distance of 51 metres to reach the surface of the sea.

- (a) Calculate the kinetic energy of the rock when it is thrown.

(2)

- (b) Calculate the potential energy lost by the rock when it reaches the surface of the sea.

(2)

- (c) (i) Find the kinetic energy of the rock when it reaches the surface of the sea.

- (ii) Hence find the speed of the rock when it reaches the surface of the sea.

(4)

- (d) State one modelling assumption which has been made.

(1)

(Total 9 marks)

Q23.

A stone, of mass 6 kg, is thrown vertically upwards with a speed of 12 m s^{-1} from a point at a height of 4 metres above ground level.

- (a) Calculate the initial kinetic energy of the stone. (2)
 - (b) (i) Show that the kinetic energy of the stone when it hits the ground is 667 J, correct to three significant figures. (2)
 - (ii) Hence find the speed of the stone when it hits the ground. (3)
 - (iii) State two modelling assumptions that you have made. (2)
- (Total 9 marks)**

Q24.

A train, of mass 60 tonnes, travels on a straight horizontal track. It has a maximum speed of 40 m s^{-1} when its engine is working at 800 kW.

- (a) Find the magnitude of the resistance force acting on the train when the train is travelling at its maximum speed. (3)
- (b) When the train is travelling at 40 m s^{-1} , the power is turned off. Assume that the resistance force is constant and is equal to that found in part (a). Also assume that this resistance force is the only horizontal force acting on the train.

Use an energy method to find how far the train travels when slowing from 40 m s^{-1} to 36 m s^{-1} .

(4)
(Total 7 marks)

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Q25.

A bungee jumper, of mass 80 kg, is attached to one end of a light elastic cord, of natural length 16 metres and modulus of elasticity 784 N. The other end of the cord is attached to a horizontal platform, which is at a height of 65 metres above the ground.

The bungee jumper steps off the platform at the point where the cord is attached and falls vertically. The bungee jumper can be modelled as a particle. Hooke's law can be assumed to apply throughout the motion and air resistance can be assumed to be negligible.

- (a) Find the length of the cord when the acceleration of the bungee jumper is zero. (3)
- (b) The cord extends by x metres beyond its natural length before the bungee jumper first comes to rest.
 - (i) Show that $x^2 - 32x - 512 = 0$. (4)

- (ii) Find the distance above the ground at which the bungee jumper first comes to rest.

(4)

(Total 11 marks)

Q26.

A slide at a water park may be modelled as a smooth plane of length 20 metres inclined at 30° to the vertical. Anne, who has a mass of 55 kg, slides down the slide. At the top of the slide, she has an initial velocity of 3 m s^{-1} down the slide.

- (a) Calculate Anne's initial kinetic energy.
- (b) By using conservation of energy, find the kinetic energy and the speed of Anne after she has travelled the 20 metres.
- (c) State one modelling assumption which you have made.

(2)

(6)

(1)

(Total 9 marks)

Q27.

A block, of mass 5 kg, is attached to one end of a length of elastic string. The other end of the string is fixed to a vertical wall. The block is placed on a horizontal surface.

The elastic string has natural length 1.2 m and modulus of elasticity 180 N. The block is pulled so that it is 2 m from the wall and is then released from rest. Whilst taut, the string remains horizontal. It may be assumed that, after the string becomes slack, it does not interfere with the movement of the block.



- (a) Calculate the elastic potential energy when the block is 2 m from the wall.
- (b) If the horizontal surface is smooth, find the speed of the block when it hits the wall.
- (c) The surface is in fact rough and the coefficient of friction between the block and the surface is μ .

(2)

(3)

Find μ if the block comes to rest just as it reaches the wall.

(7)

(Total 12 marks)

Q28.

A ball is thrown vertically upwards from ground level with an initial speed of 15 m s^{-1} . The ball has a mass of 0.6 kg . Assume that the only force acting on the ball after it is thrown is its weight.

- (a) Calculate the initial kinetic energy of the ball. (2)
 - (b) By using conservation of energy, find the maximum height above ground level reached by the ball. (3)
 - (c) By using conservation of energy, find the kinetic energy and the speed of the ball when it is at a height of 3 m above ground level. (4)
- (Total 9 marks)**

Q29.

A ball is thrown vertically upwards from ground level with an initial speed of 15 m s^{-1} . The ball has a mass of 0.6 kg . Assume that the only force acting on the ball after it is thrown is its weight.

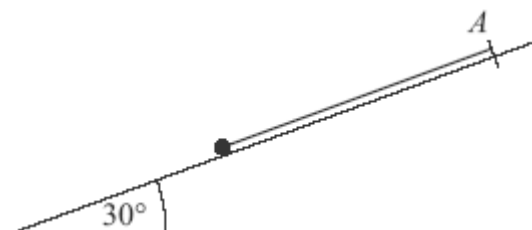
- (a) Calculate the initial kinetic energy of the ball. (2)
 - (b) By using conservation of energy, find the maximum height above ground level reached by the ball. (3)
 - (c) By using conservation of energy, find the kinetic energy and the speed of the ball when it is at a height of 3 m above ground level. (4)
 - (d) State one modelling assumption which has been made. (1)
- (Total 10 marks)**

Q30.

A light elastic string has one end attached to a point A fixed on a smooth plane inclined at 30° to the horizontal. The other end of the string is attached to a particle of mass 6 kg . The elastic string has natural length 4 metres and modulus of elasticity 300 newtons .

The particle is pulled down the plane in the direction of the line of greatest slope through A .

The particle is released from rest when it is 5.5 metres from A .



- (a) Calculate the elastic potential energy of the string when the particle is 5.5 metres from the point A . (2)
- (b) Show that the speed of the particle when the string becomes slack is 3.66 m s^{-1} , correct to three significant figures. (5)
- (c) Show that the particle will **not** reach point A in the subsequent motion. (3)

(Total 10 marks)

Q31.

- (a) Hooke's law states that the tension in a stretched string of natural length l and modulus of elasticity λ is $\frac{\lambda x}{l}$ when its extension is x .

Using this formula, prove that the work done in stretching a string from an

unstretched position to a position in which its extension is e is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg, is attached to one end of a light elastic string of natural length 0.6 metres and modulus of elasticity 150 N. The other end of the string is fixed to a point O .

- (i) Find the extension of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (ii) The particle is pulled down and held at the point P , which is 2 metres vertically below O .

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Show that the elastic potential energy of the string when the particle is in this position is 245 J.

(2)

- (iii) The particle is released from rest at the point P . Find the speed of the particle when it reaches O .

(4)

(Total 11 marks)

Q32.

- (a) Hooke's law states that the tension in a stretched string of natural length l and modulus of elasticity λ is $\frac{\lambda x}{l}$ when its extension is x .

Using this formula, prove that the work done in stretching a string from an

unstretched position to a position in which its extension is e is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg, is attached to one end of a light elastic string of natural length 0.6 metres and modulus of elasticity 150 N. The other end of the string is fixed to a point O .

- (i) Find the extension of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (ii) The particle is pulled down and held at the point P , which is 0.9 metres vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 11.25 J.

(2)

- (iii) The particle is released from rest at the point P . In the subsequent motion, the particle has speed $v \text{ m s}^{-1}$ when it is x metres **above** P .

Show that, while the string is taut,

$$v^2 = 10.4x - 50x^2$$

(7)

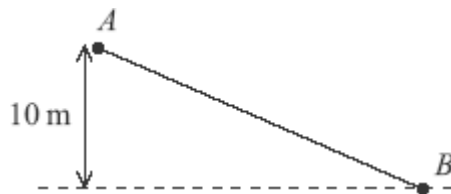
- (iv) Find the value of x when the particle comes to rest for the first time after being released, given that the string is still taut.

(2)

(Total 16 marks)

Q33.

A child, of mass 35 kg, slides down a slide in a water park. The child, starting from rest, slides from the point A to the point B , which is 10 metres vertically below the level of A , as shown in the diagram.



- (a) In a simple model, all resistance forces are ignored.

Use an energy method to find the speed of the child at B .

(3)

- (b) State one resistance force that has been ignored in answering part (a).

(1)

- (c) In fact, when the child slides down the slide, she reaches B with a speed of 12 m s^{-1} .

Given that the slide is 20 metres long and the sum of the resistance forces has a

constant magnitude of F newtons, use an energy method to find the value of F .

(4)

(Total 8 marks)

Q34.

An elastic string of natural length 1.5 metres has one end attached to a fixed point O . A particle of mass 4 kg is attached to the other end of the string. The particle is released from rest at O .

- (a) Find the kinetic energy of the particle when the string becomes taut.

(2)

- (b) The particle first comes to rest when it is 3.5 metres below O .

Show that the modulus of elasticity of the string is 103 N, correct to three significant figures.

(4)

- (c) Find the speed of the particle when it is 2.7 metres below O .

(5)

(Total 11 marks)

Q35.

A hot air balloon moves vertically upwards with a constant velocity. When the balloon is at a height of 30 metres above ground level, a box of mass 5 kg is released from the balloon.

After the box is released, it initially moves vertically upwards with speed 10 m s^{-1} .

- (a) Find the initial kinetic energy of the box.

(2)

- (b) Show that the kinetic energy of the box when it hits the ground is 1720 J.

(3)

- (c) Hence find the speed of the box when it hits the ground.

(3)

- (d) State **two** modelling assumptions which you have made.

(2)

(Total 10 marks)

Q36.

An elastic string has one end attached to a point O , fixed on a horizontal table. The other end of the string is attached to a particle of mass 5 kilograms. The elastic string has natural length 2 metres and modulus of elasticity 200 newtons. The particle is pulled so that it is 2.5 metres from the point O and it is then released from rest on the table.

- (a) Calculate the elastic potential energy when the particle is 2.5 m from the point O .

(2)

- (b) If the table is smooth, show that the speed of the particle when the string becomes

slack is $\sqrt{5} \text{ m s}^{-1}$.

(3)

- (c) The table is, in fact, rough and the coefficient of friction between the particle and the table is 0.4.

Find the speed of the particle when the string becomes slack.

(7)

(Total 12 marks)

Q37.

A particle, of mass 10 kg, is attached to one end of a light elastic string of natural length 0.4 metres and modulus of elasticity 100 N. The other end of the string is fixed to the point O .

- (a) Find the length of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (b) The particle is pulled down and held at a point P , which is 1 metre vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 45 J.

(2)

- (c) The particle is released from rest at the point P . In the subsequent motion, the particle has speed $v \text{ m s}^{-1}$ when it is x metres **below** O .

- (i) Show that, while the string is taut,

$$v^2 = 39.6x - 25x^2 - 14.6$$

(7)

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- (ii) Find the value of x when the particle comes to rest for the first time after being released, given that the string is still taut.

(3)

(Total 14 marks)

Q38.

A stone, of mass 0.4 kg, is thrown vertically upwards with a speed of 8 m s^{-1} from a point at a height of 6 metres above ground level.

- (a) Calculate the initial kinetic energy of the stone.

(2)

- (b) (i) Show that the kinetic energy of the stone when it hits the ground is 36.3 J, correct to three significant figures.

(2)

- (ii) Hence find the speed at which the stone hits the ground.

(3)

- (iii) State one assumption that you have made.

(1)

(Total 8 marks)

Q39.

A block of mass 2 kg is placed on a horizontal surface. An elastic string has natural length 0.5 metres and modulus of elasticity 30 newtons. One end of the string is fixed to the surface at the point O and the other end is attached to the block. The block is pulled along the surface away from O until it is at the point P , where the length of OP is 1.8 metres. The block can be modelled as a particle.

- (a) Calculate the elastic potential energy in the string when the block is at P .

(2)

- (b) Assume that the horizontal surface is smooth.

- (i) The block is then released from P and moves towards O . Show that, when the block has moved 0.5 metres, its speed is 5.61 m s^{-1} , correct to three significant figures.

(5)

- (ii) Find the speed of the block when it reaches O .

(3)

- (c) Assume that the horizontal surface is rough and that the coefficient of friction between the surface and the block is 0.1. Find the speed of the block when it reaches O .

(5)

(Total 15 marks)

Q40.

A ball of mass 0.6 kg is thrown vertically upwards from ground level with an initial speed of 14 m s^{-1} . ©2025 Exam Papers Practice. All rights reserved.

- (a) Calculate the initial kinetic energy of the ball.

(2)

- (b) Assuming that no resistance forces act on the ball, use an energy method to find the maximum height reached by the ball.

(3)

- (c) An experiment is conducted to confirm the maximum height for the ball calculated in part (b). In this experiment the ball rises to a height of only 8 metres.

- (i) Find the work done against the air resistance force that acts on the ball as it moves.

(3)

- (ii) Assuming that the air resistance force is constant, find its magnitude.

(2)

- (d) Explain why it is **not** realistic to model the air resistance as a constant force.

(1)
(Total 11 marks)

Q41.

A light spring of natural length 0.4 metres and modulus of elasticity λ newtons lies on a smooth horizontal table. One end of the spring is attached to the table at a point O and the other end is attached to a particle. The particle is held in equilibrium on the table by a horizontal force of magnitude 12.5 newtons. In this position the length of the spring is 0.5 metres.

- (a) Show that $\lambda = 50$.

(3)

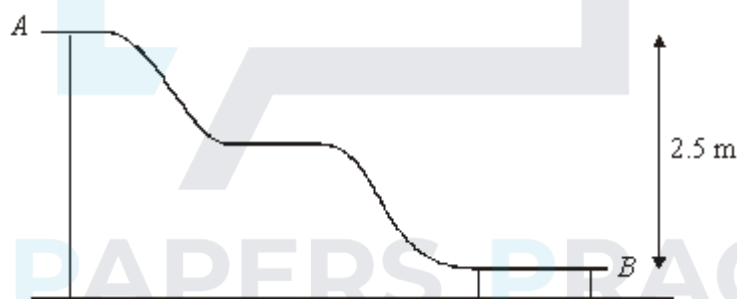
- (b) The horizontal force is removed and the particle, which is of mass 0.2 kg, is released from rest and moves towards O . Calculate its speed when the spring reaches its natural length.

(5)

(Total 8 marks)

Q42.

The diagram below shows the cross-section of a slide in a children's playground. Jan, of mass 28 kg, sits at rest on the slide at A .



Jan's friend gives her a push which causes her to start moving with speed 1 m s^{-1} . She then slides down the slide to B , which is 2.5 metres below the level of A .

- (a) In a simple model, Jan's motion is not affected by any forces of resistance.

- (i) By using energy, show that Jan's speed when she reaches B is 7.07 m s^{-1} , correct to three significant figures.

(4)

- (ii) It takes T seconds for Jan to move from A to B . Sketch a velocity-time graph for Jan's motion as she travels from A to B .

(3)

- (b) In a more refined model, Jan's motion is assumed to be affected by resistance forces. Given that the work done by the resistance forces while Jan travels from A to B is 350 J, calculate Jan's speed when she reaches B .

(4)

(Total 11 marks)

Q43.

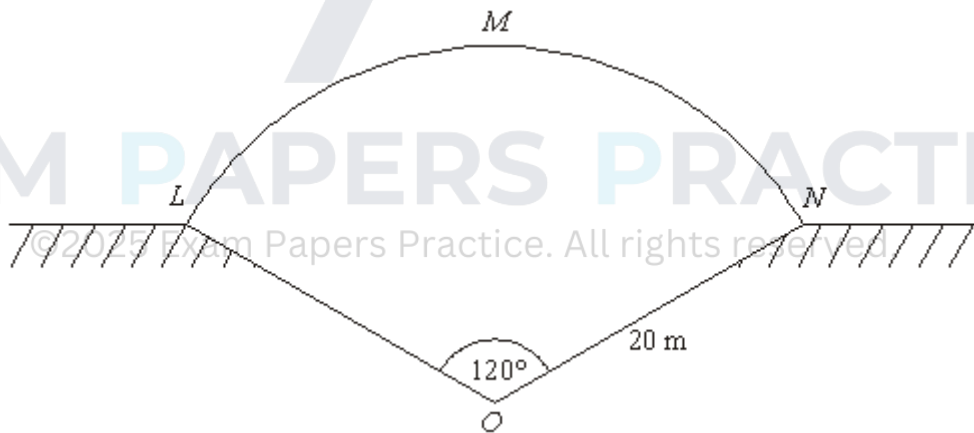
Alina, who has mass 50 kg, is attached to one end of an elastic cord. The cord has natural length 20 metres and stiffness $k \text{ N m}^{-1}$. The other end of the cord is attached to a bridge. Alina steps off the bridge at the point where the cord is attached and falls vertically. Throughout the subsequent motion, Alina can be modelled as a particle.

- (a) Find Alina's speed when she passes through the point 20 metres vertically below the bridge. (2)
- (b) The cord stretches to a total length of 32 metres before Alina comes momentarily to rest.
 - (i) Show that the elastic potential energy of the cord at the instant when she is momentarily at rest is 15 680 J. (2)
 - (ii) Hence find the stiffness of the cord. (3)

(Total 7 marks)

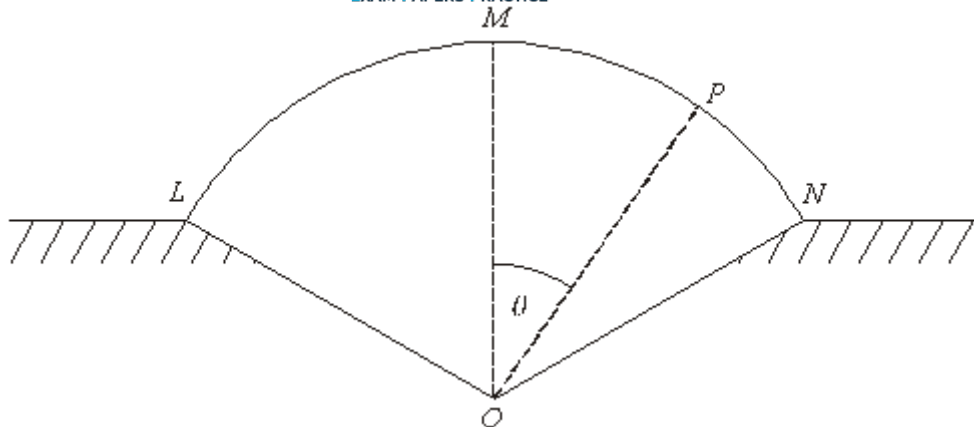
Q44.

A cyclist approaches a hump-backed bridge which is in the form of a circular arc LMN . The radius of the arc is 20 m and its centre is O . The angle between OL and ON is 120° . The points L and N are on the same horizontal level and M is the highest point of the bridge, as shown in the diagram.



The total mass of the cycle and cyclist is 60 kg. They may be modelled as a single particle. Assume that there is no resistance to the motion.

- (a) Calculate the gain in potential energy as the cyclist moves from L to M . (3)
- (b) At M the speed of the cyclist is 6 m s^{-1} . He begins to freewheel from M towards N . The point P is on the bridge between M and N and angle $MOP = \theta$, as shown in the diagram.



When the cyclist is at the point P his speed is v .

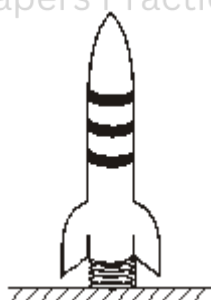
- (i) Show that $v^2 = 428 - 392 \cos \theta$. (4)
 - (ii) Find, in terms of θ , the normal reaction between the bridge and the cycle at P . (4)
 - (iii) Determine whether or not the cycle remains in contact with the bridge throughout the descent from M to N . (3)
- (Total 14 marks)

Q45.

A child's pop-up toy rocket is of mass 20 grams. It is operated by pushing the rocket vertically downwards to compress a light spring underneath the rocket. The rocket is then released from rest.

Initially, the spring is compressed by 3 cm to the position shown in the diagram. The stiffness of the spring is 50 N m^{-1} .

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- (a) Calculate the elastic potential energy of the compressed spring. (2)
- (b) When released, the rocket moves vertically upwards. Show that the speed of the rocket when the spring first reaches its natural length is approximately 1.29 m s^{-1} . (4)
- (c) Determine the height above the initial position, shown in the diagram, at which the rocket comes momentarily to rest. (3)

Q46.



The diagram shows a vertical cross section of a new adventure slide at a theme park. It consists of three sections AB , BC and CD .

Section AB is smooth and vertical and has length r .

Section BC is smooth and forms a quarter of a circle. This circle has centre O and radius r . The radius OB is horizontal and OC is vertical.

Section CD is rough, straight and horizontal. It is of length $4r$.

Steve, who has mass m , starts from rest at A and reaches speed u at the point B . He remains in contact with the surface until he reaches D .

It can be assumed that Steve can be modelled as a particle throughout the motion.

- (a) Find u^2 in terms of g and r . (2)

- (b) Steve reaches the point P between B and C where angle $POB = \theta$, as shown in the diagram. His speed at P is v .
 - (i) Show that $v^2 = 2gr(1 + \sin \theta)$. (4)

 - (ii) Draw a diagram showing the forces acting on Steve when he is at the point P . (1)

 - (iii) Find an expression for the normal reaction, R , on Steve when he is at the point P . Give your answer in terms of m , g and θ . (4)

- (c) Show that, as Steve crosses C , there is a reduction in the normal reaction of

magnitude 4 mg .

(2)

- (d) Between C and D , Steve decelerates uniformly and comes to rest at the point D .

Find his retardation.

(3)

(Total 16 marks)

Q47.

A bungee jumper, of mass 70 kg , is attached to one end of a light elastic cord of natural length 14 metres and stiffness 196 N m^{-1} . The other end of the cord is attached to a bridge, approximately 40 metres above a river.

The bungee jumper steps off the bridge at the point where the cord is attached and falls vertically. The bungee jumper can be modelled as a particle throughout the motion. Hooke's law can be assumed to apply throughout the motion.

- (a) Find the speed of the bungee jumper at the instant the cord first becomes taut.

(2)

- (b) The cord extends by $x \text{ metres}$ beyond its natural length before the bungee jumper first comes instantaneously to rest.

- (i) Show that $x^2 - 7x - 98 = 0$.

(4)

- (ii) Hence find the value of x .

(2)

- (iii) Calculate the deceleration experienced by the bungee jumper at this point.

(4)

(Total 12 marks)

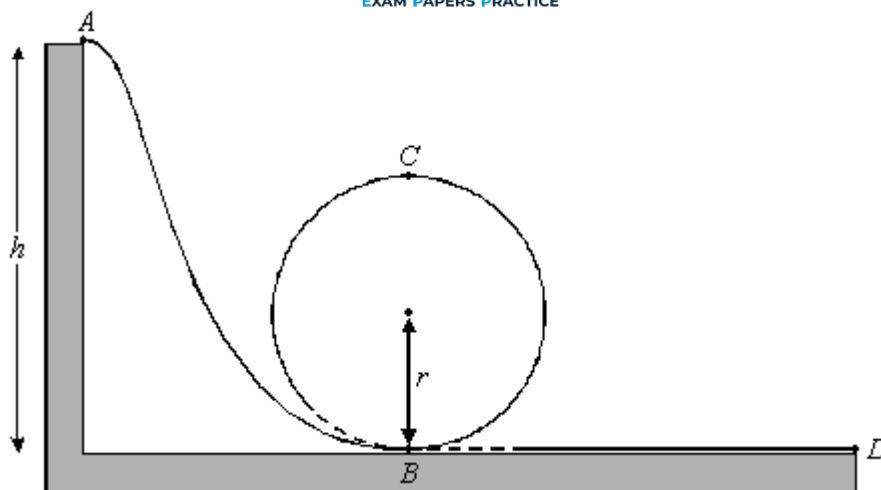
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Q48.

Maria is modelling the motion of a toy car along a "loop the loop" track.

The track, $ABCD$, can be modelled as a continuous smooth surface contained in a vertical plane. The highest point on the track is A , which is a distance h above the floor. The loop is a circle of radius r with BC as a diameter, and with C vertically above B . The track ends at the point D , as shown in the diagram.

Maria is trying to determine a connection between h and r in the case where the car stays only just in contact with the track at C .



Maria models the car as a particle of mass m , which starts from rest at A . In the case where the car stays only just in contact with the track at C , it has speed u at B and speed v at C .

- (a) By considering the forces on the car at C , show that $v^2 = rg$. (3)
 - (b) Find an expression for u^2 in terms of g and r . (4)
 - (c) Show that $h = kr$, where k is a constant to be determined. (3)
 - (d) Suggest one improvement that could be made to the model in order to refine the solution. (1)
- (Total 11 marks)

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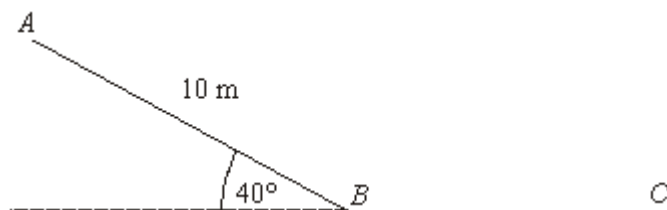
A small sphere, of mass 5 kg, is attached to one end of a light elastic string. The other end of the string is fixed to a smooth horizontal surface at the point O . The string has modulus of elasticity 40 N and natural length 2 metres.

The sphere is released from rest at the point P , which is on the horizontal surface 5 metres from O .

- (a) Show that the elastic potential energy of the string is 90 J when the sphere is released. (2)
 - (b) Show that the speed of the sphere is 6 m s^{-1} when the string becomes slack. (3)
 - (c) Find the speed of the sphere when it is 3 metres from O . (5)
- (Total 10 marks)

Q50.

The diagram shows a simple slide made of two sections AB and BC . Section AB has length 10 metres and is at an angle of 40° to the horizontal. Section BC is horizontal.



A child uses the slide, pushing himself off so that at A his speed is 2 m s^{-1} . When he reaches B his speed is 6 m s^{-1} . The mass of the child is 35 kg.

- (a) (i) Calculate the kinetic energy of the child at A . (2)
- (ii) Calculate the gain in kinetic energy as the child moves from A to B . (3)
- (b) Show that the loss in potential energy as the child moves from A to B is 2200 joules, correct to three significant figures. (2)
- (c) As the child moves from A to B , a resistance force acts on him. Assume that this force has a constant magnitude F newtons.
Find F . (3)
- (d) Assume that the same resistance force acts on the child as he moves on BC and that he stops at C . Find the length of BC . (3)

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(Total 13 marks)

Q51.

An elastic string has natural length 2 metres and modulus of elasticity 30 N. One end of the string is fixed at the point O . A particle of mass 0.15 kg is attached to the other end of the string. The particle is pulled down until it is 2.8 metres directly below O and released from rest.

- (a) Calculate the elastic potential energy of the string just before the particle is released. (2)
- (b) After the particle has been released, it moves upwards.
 - (i) Show that the speed of the particle as it passes O is 3.02 m s^{-1} , correct to three significant figures. (4)

- (ii) By considering the maximum height of the particle above O , show that the string does not become taut as the particle moves upwards.

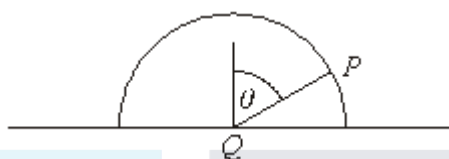
(4)

(Total 10 marks)

Q52.

A smooth hemisphere of radius a and centre Q lies with its plane face fixed to a horizontal surface. A particle of mass m can move freely on the surface of the hemisphere.

Initially, the particle is at rest at the highest point of the hemisphere. It is set in motion along the surface of the hemisphere. The line QP makes an angle of θ with the upward vertical.



- (a) Show that, when the particle is at P , its speed is

$$\sqrt{2ga(1 - \cos \theta)}$$

(3)

- (b) Find the value of $\cos \theta$ if the particle leaves the surface of the hemisphere when it reaches P .

(4)

(Total 7 marks)

Q53.

A diver has mass 65 kg. She dives from a fixed diving board, which is 6 metres above the level of the water in the pool. When the diver leaves the board, she is travelling vertically upwards and has speed 2 m s^{-1} .

Model the diver as a particle. Assume that there are no resistance forces acting on the diver as she moves through the air and that she does not hit the board on the way down.

- (a) (i) Calculate the kinetic energy of the diver when she leaves the board.
- (ii) By using an energy method, calculate the maximum height of the diver above the diving board.
- (b) (i) Find the kinetic energy of the diver when she hits the water.
- (ii) Hence calculate the speed of the diver when she hits the water.

(2)

(2)

(3)

(2)

(Total 9 marks)

Q54.

An elastic string has natural length 2 metres and modulus of elasticity λ newtons. One end of the string is fixed at the point O , and a particle of mass 20 kg is attached to the other end of the string.

- (a) When in equilibrium the particle is 2.7 metres below O . Show that $\lambda = 560$.

(3)

- (b) The particle is now held at O and released from rest. The maximum length of the string in the subsequent motion is L .

- (i) Show that L satisfies the equation

$$5L^2 - 27L + 20 = 0$$

(5)

- (ii) Find the maximum length of the string.

(3)

(Total 11 marks)

Q55.

An elastic rope has natural length 4 metres and modulus of elasticity 80 N. A particle, of mass 2 kg, is attached to one end of the rope, and the other end is fixed at the point A . The particle is released from rest at A and falls vertically.

- (a) When the rope just becomes taut, find:

- (i) the kinetic energy of the particle;

(2)

- (ii) the speed of the particle.

(3)

- (b) (i) The maximum extension of the rope during the motion is x metres. Show that x satisfies the equation

$$10x^2 - 19.6x - 78.4 = 0$$

(4)

- (ii) Hence find the maximum length of the rope.

(3)

- (c) State clearly **one** important assumption that you have made.

(1)

(Total 13 marks)

Q56.

A ball is projected vertically upwards, from ground level, with an initial speed of 18 m s^{-1} . The ball has a mass of 0.3 kg. Assume that the force of gravity is the only force acting on the ball after it is projected.

- (a) Calculate the initial kinetic energy of the ball. (2)
- (b) By using conservation of energy, find the maximum height of the ball above ground level. (2)
- (c) Find the kinetic energy and the speed of the ball when it is at a height of 2 metres above ground level. (5)
- (Total 9 marks)**

Q57.

An elastic string has modulus of elasticity 12 N and natural length 0.5 metres. A particle of mass 0.5 kg is attached to one end of the string. The other end of the string is attached to a fixed point P . The particle is pulled down until it is 1.5 metres below P .

- (a) Calculate the elastic potential energy of the string when the particle is 1.5 metres below P . (2)
- (b) The particle is released.
- (i) Show that the kinetic energy of the particle is 7.1 J, when the string becomes slack. (2)
- (ii) Find the kinetic energy of the particle when it is 0.5 metres above P . (2)
- (iii) Find the maximum height of the particle above P . (7)

(Total 13 marks)

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Q58.

A ball has mass 0.5 kg and is released from rest at a height of 6 metres above ground level.

- (a) Assume that no resistance force acts on the ball as it falls.
- (i) Find the kinetic energy of the ball when it has fallen 3 metres. (2)
- (ii) Use an energy method to find the speed of the ball when it hits the ground. (3)
- (b) Assume that a constant resistance force acts on the ball as it falls and that the ball hits the ground travelling at 2 m s^{-1} . Use an energy method to find the magnitude of the resistance force. (4)

(Total 9 marks)